Problem Set #10 – Fundamental Algorithm Techniques

Aibek Zhazykbek

Problem 1 — Complexity Classes

We classify each line into P, NP, NP-hard, NP-complete or undecidable.

1. **find max, linear search, shortest path in unweighted graph, matrix multiplication** All are solvable in polynomial time.

   $\Rightarrow$ **Class: P**

2. **sorting, Dijkstra, BFS, DFS, mergesort, quicksort** All run in polynomial time.

   $\Rightarrow$ **Class: P**

3. **sudoku** Sudoku is known to be NP-complete (as a decision problem).

   $\Rightarrow$ **NP-complete**

4. **3-coloring, scheduling with conflicts** Both are NP-complete.

   $\Rightarrow$ **NP-complete**

5. **TSP, Hamiltonian Cycle, Clique** These are classical NP-complete optimization/decision problems.

   $\Rightarrow$ **NP-complete or NP-hard**

6. **Cryptography, factoring** Factoring is in NP but not known to be in P or NP-complete. Used in cryptography.

   $\Rightarrow$ **NP, not known to be P or NP-complete**

7. **Halting Problem, Busy Beaver** Both are undecidable.

   $\Rightarrow$ **Undecidable**

Problem 2 — Bayes Theorem

Prior disease probability:

$$P(D) = 0.001$$

Test accuracy:

$$P(\text{pos} \mid D) = 0.99, \qquad P(\text{pos} \mid \neg D) = 0.01$$

Bayes formula:

$$P(D \mid \text{pos}) = \frac{P(\text{pos} \mid D)\,P(D)}{P(\text{pos} \mid D)P(D) + P(\text{pos} \mid \neg D)P(\neg D)}$$

Compute:

$$= \frac{0.99 \cdot 0.001}{0.99 \cdot 0.001 + 0.01 \cdot 0.999} = 0.0901 \approx 9\%.$$

The reason: false positives dominate when the disease is extremely rare.

Problem 3 — Shannon Entropy

Shannon entropy:

$$H(X) = -\sum p_i \log_2 p_i$$

For a coin:

$$H = -p\log_2 p - (1-p)\log_2(1-p)$$

- **Fair coin** ($p = 0.5$):

$$H = 1 \text{ bit}$$

- **99% heads coin**: $p = 0.99$

$$H = -0.99\log_2 0.99 - 0.01\log_2 0.01 = 0.08 \text{ bits}$$

- **1% heads coin** is symmetric:

$$H = 0.08 \text{ bits}$$

Interpretation: A fair coin is fully uncertain → needs 1 bit. A 99% biased coin is almost predictable → only 0.08 bits of information.